

Table 2: **Probing Results (Layer 24)**. Values indicate the percentage of prompts classified as *Evaluation*. Standard training conflates format with context, leading to high false positives on *Bench-Deploy*. Paired training restores correct classification.

Training Regime	Intended Evaluation		Intended Deployment	
	Bench-Eval	Casual-Eval	Casual-Deploy	Bench-Deploy
Standard (Single)	98.8%	0.0%	0.0%	94.5% (FP)
Paired (Decorrelated)	98.5%	100.0%	0.0%	7.2%

Paired Training Success. When format is decorrelated from context during training, the probe successfully learns the underlying concept. As shown in Table 2 (bottom row), the paired probe correctly identifies *Bench-Deploy* as deployment (7.2% error vs 94.5%) and recognizes *Casual-Eval* as evaluation (100%). Furthermore, we evaluated these probes on AI Safety datasets (AdvBench, HarmBench) and found that neither standard nor paired probes flag safety prompts as "Evaluation" (see Appendix A.5).

5 REASON FOR FAILURE

Diagnosis. The observed collapse indicates that probe-based diagnostics exploit surface-level structural features rather than encoding an abstract notion of evaluation context. Benchmark-canonical layouts introduce strong, linearly separable cues in representation space, causing probes to conflate "being evaluated" with "looking like a benchmark."

Why this is not a trivial confound. Prompt length is explicitly matched across evaluation and deployment datasets, ruling out length-based shortcuts. Genre and formality are manipulated independently from structure, yet do not induce evaluation-like predictions. We explicitly audit and remove *rewrite-induced answer leakage*, where benchmark-format rewrites of deployment prompts inadvertently include model-generated answer content (e.g., pre-filled options or solution scaffolds). Ablations show that removing such leakage does not explain the observed effect and in some cases increases probe confidence. The failure is consistent across layers, indicating a pervasive structural bias rather than a localized artifact.

Boundary conditions. We do not claim that evaluation awareness is absent from model representations. Rather, our results show that under standard benchmark-correlated training setups, linear probes fail to isolate such signals. Only when format is explicitly decorrelated from context during training do probes exhibit format-invariant behavior.

6 DISCUSSION

Implications. Our results suggest that probe-based monitoring of evaluation awareness can induce false confidence. Diagnostics sensitive to prompt format rather than usage context risk mischaracterizing model behavior, and governance mechanisms relying on probe accuracy may systematically overestimate situational awareness. Benchmark-canonical structure must be explicitly controlled for in experimental design, not bundled with context by default. Without such control, probe accuracy may reflect sensitivity to layout rather than higher-level contextual reasoning.

Limitations. Our study uses only linear probes and a single backbone (Llama-3.1-8B-Instruct), and does not rule out non-linear or distributed representations of evaluation awareness. Some dataset instantiations rely on proxy constructions. We expect the central confound—benchmark structure correlating with evaluation context—to persist broadly due to common benchmark construction practices.

REPRODUCIBILITY STATEMENT

All code and constructed datasets are publicly available at <https://github.com/viliana-dev/eval-awareness-format>. The repository includes scripts for the full pipeline: dataset downloading, query routing, format rewrites, histogram-based length matching, activation extraction, probe training, and all evaluation protocols reported in this paper. Pre-processed datasets for the 2×2 matrix are hosted at <https://huggingface.co/datasets/viliana-dev/eval-awareness-2x2>.

All source datasets are public: MMLU (Hendrycks et al., 2021), ARC (Clark et al., 2018), GSM8K (Cobbe et al., 2021), ShareGPT¹, and MuTual (Cui et al., 2020). Benchmark-format rewrites were generated using DeepSeek-V3; exact prompt templates are provided in Appendix A.6.

All experiments use `meta-llama/Llama-3.1-8B-Instruct` with `bfloat16` inference on a single NVIDIA A100 80 GB. Probes are trained in `float32`. A fixed random seed of 42 is used throughout all scripts. Implementation details—layer indices, sample sizes ($n=1076$), regularization, and cross-validation settings—are reported in Appendix A.4.

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A APPENDIX: DATASETS, CONTROLS, AND ADDITIONAL EXPERIMENTS

A.1 DATASETS IN THE 2×2 MATRIX

We use a controlled dataset matrix crossing **usage context** (evaluation vs. deployment) and **genre** (benchmark-style vs. conversational), while treating **format** (benchmark-canonical structured vs. free-form) as a separate variable.

¹<https://huggingface.co/datasets/theblackcat102/sharegpt-english>

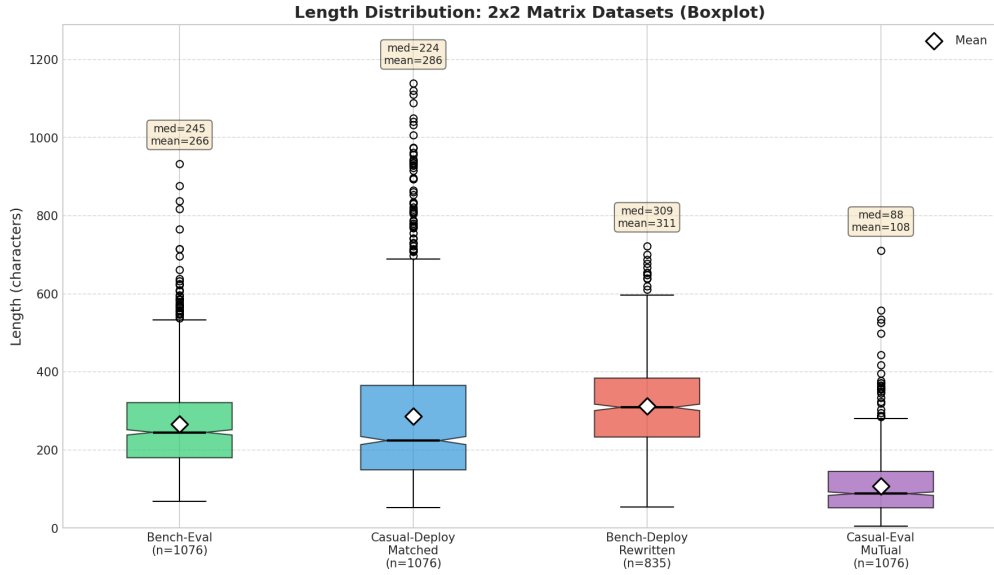


Figure 2: Length distributions (in characters) across the four datasets. *Casual-Deploy* is histogram-matched to *Bench-Eval*. *Bench-Deploy* is slightly longer due to formatting overhead, while *Casual-Eval* (1st turn) is naturally shorter.

Bench-Eval (Evaluation; benchmark-style; structured). Canonical benchmark prompts (e.g., MCQ and structured word-problem formats) drawn from MMLU, ARC, and GSM8K.

Casual-Deploy (Deployment; conversational; free-form). Deployment-style user prompts sampled from ShareGPT. For comparability with Bench-Eval, we use only the first user turn and apply histogram matching for length control (see Section A.2).

Casual-Eval (Evaluation; conversational; free-form proxy). We instantiate this condition with MuTual as a conversational evaluation proxy. For tighter control over genre and input structure, we extract only the first dialogue turn from each example; we treat this as an approximation rather than a faithful instantiation of the original task.

Bench-Deploy (Deployment; benchmark-style; structured proxy via rewrites). Deployment-origin prompts from Casual-Deploy rewritten into benchmark-canonical structured templates (MCQ or structured word problems), while avoiding explicit evaluation framing. Since not all deployment prompts can be sensibly forced into benchmark templates, we first route prompts into coarse task categories and rewrite only compatible prompts.

A.2 LENGTH CONTROL

Motivation. Initially, the datasets exhibited significant length disparities, creating a potential shortcut for probes. *Bench-Eval* prompts had a median length of 248 characters, while the raw *Casual-Deploy* (ShareGPT 1st turn) prompts were much shorter (median 113 characters).

Histogram Matching. To eliminate length as a confounding variable in the primary training contrast (*Bench-Eval* vs. *Casual-Deploy*), we performed histogram matching. We calculated the length distribution of *Bench-Eval* using 30 percentile-based character-length bins and sampled the exact same number of examples from *Casual-Deploy* for each bin. This process aligned the distributions as closely as possible, yielding equal sample sizes ($n = 1076$) and comparable statistics. Figure 2 visualizes the resulting length distributions.